SRADHA PANIGRAHI

Electrical & Electronics Engineering

Berhampur, Odisha, India | +91 6371443909 | sradhapanigrahi539@gmail.com | linkedin.com/in/sradha-panigrahi-ab732a290

PROFESSIONAL SUMMARY

Electrical & Electronics Engineering undergraduate at NIST University with experience in power systems, circuit simulation, embedded systems, and IoT. Familiar with MATLAB/Simulink, ETAP, COMSOL Multiphysics, Arduino, ESP32/ESP8266, and Python. Completed internships at IIT Bhubaneswar and TP Southern Odisha Distribution Limited (TPSODL), with project experience in designing ESP and solar pv system.

EDUCATION

Bachelor of Technology (B.Tech), Electrical & Electronics Engineering — NIST University, Berhampur, Odisha | *2023 – Present*

Senior Secondary (+2) — Sankalp Senior Secondary School, Berhampur, Odisha | *2020 – 2022*

Secondary School — K.C Public School, Berhampur, Odisha | *2008 – 2020*

TECHNICAL SKILLS

Programming: Python, Assembly Language ,SQL, C
Simulation & Design: MATLAB, Simulink, LabVIEW, ETAP, COMSOL Multiphysics
Embedded Systems: Arduino, ESP32/ESP8266, Raspberry Pi, 8051, 8086, 8255
IoT: Sensor Integration, Home Automation, Actuators, MQTT
Power Systems: Power Generation, Transmission & Distribution, Switchgear & Protection, Renewable Energy Systems

INTERNSHIP EXPERIENCE

Research Intern — IIT Bhubaneswar, Bhubaneswar, Odisha | *May 2026 – Jul 2026*

- Applied COMSOL Multiphysics to design and simulate an Electrostatic Precipitator (ESP), covering geometry modeling, meshing, and electric-field analysis .

Power Systems Intern — TP Southern Odisha Distribution Limited (TPSODL), Odisha | *Jun 2025 – Jul 2025*

- Analyzed power distribution architectures with hands-on exposure to substation design and grid management.
- Studied predictive maintenance techniques for transformers and circuit breakers.
- Evaluated performance metrics of high-voltage electrical assets to support operational-efficiency improvements.

C Programming Intern — NIST University, Berhampur, Odisha | *May 2024 – Jun 2024*

- Built foundational skills in structured programming and problem-solving using C.

PROJECTS

Electrostatic Precipitator (ESP) Design & Simulation | *COMSOL Multiphysics*

- Designed and simulated a 2D ESP integrating electrostatics and particle-tracing physics to analyze particulate-matter collection.
- Performed geometry modeling, boundary-condition setup, meshing, and simulation to evaluate electric-field distribution and optimize collection efficiency.

PORTA-MP: Portable AI-Enhanced Microplastic Detector | *Raspberry Pi , Python*

- Engineered a low-cost, field-ready device using Raspberry Pi 5, UV/Blue LED arrays, and optical emission filters to detect microplastics in water via Nile Red fluorescence staining without complex lab infrastructure.
- For real-time particle counting and size estimation, with a Python-based GUI dashboard for instant data visualization and CSV export for regulatory reporting.

Knocking Pattern Based Smart Door Lock | *ESP32, IoT, Sensor Integration*

- Built a keyless smart door lock using ESP32 and a piezo sensor to detect and authenticate custom knock patterns in real time, triggering relay-based solenoid unlocking.
- Implemented multi-user pattern customization with OLED status display, enabling a low-cost, scalable access control system.

Grid-Connected PV Microgrid Modeling with MPPT Controller | *MATLAB, Simulink*

- Modeled a 6.84 kW grid-connected solar PV array and integrated a lithium-ion BESS with MPPT control to stabilize output and reduce power fluctuations.
- Applied a moving-average smoothing algorithm and voltage-sag mitigation to improve reliability under variable weather conditions.

ACHIEVEMENTS

- 2nd Place, Circuit Design Competition, NIST University — designed a functional electrical circuit under an accelerated deadline, demonstrating strong analytical and troubleshooting skills.
- Paper Presentation Winner, NIST University — presented a paper on smart irrigation systems.

CERTIFICATIONS

- NPTEL: Foundation of Cloud IoT Edge ML
- Internshala: UI/UX Designing
- NPTEL: Smart Grid-Basics to Advanced Technologies